

Import Data

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [2]: data = pd.read_csv(r"D:\Downloads\Udemy-PY\ai_financial_market_daily_realistic_synt
```

```
In [3]: data
```

```
Out[3]:
```

	Date	Company	R&D_Spending_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth
0	2015-01-01	OpenAI	5.92	0.63	-36
1	2015-01-02	OpenAI	5.41	1.81	80
2	2015-01-03	OpenAI	4.50	0.61	-38
3	2015-01-04	OpenAI	5.45	0.95	-5
4	2015-01-05	OpenAI	3.40	1.48	48
...	...	...	...	...	...
10954	2024-12-27	Meta	100.19	103.54	417
10955	2024-12-28	Meta	99.12	102.37	411
10956	2024-12-29	Meta	98.95	103.11	415
10957	2024-12-30	Meta	100.74	103.21	416
10958	2024-12-31	Meta	100.08	103.41	417

10959 rows × 7 columns



```
In [4]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10959 entries, 0 to 10958
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Date                                  10959 non-null  object
1   Company                              10959 non-null  object
2   R&D_Spending_USD_Mn                 10959 non-null  float64
3   AI_Revenue_USD_Mn                   10959 non-null  float64
4   AI_Revenue_Growth_%                 10959 non-null  float64
5   Event                                233 non-null    object
6   Stock_Impact_%                      10959 non-null  float64
dtypes: float64(4), object(3)
memory usage: 599.4+ KB
```

```
In [5]: data['Date'] = pd.to_datetime(data['Date'])
```

```
In [6]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10959 entries, 0 to 10958
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Date                                  10959 non-null  datetime64[ns]
1   Company                              10959 non-null  object
2   R&D_Spending_USD_Mn                 10959 non-null  float64
3   AI_Revenue_USD_Mn                   10959 non-null  float64
4   AI_Revenue_Growth_%                 10959 non-null  float64
5   Event                                233 non-null    object
6   Stock_Impact_%                      10959 non-null  float64
dtypes: datetime64[ns](1), float64(4), object(2)
memory usage: 599.4+ KB
```

```
In [7]: data['Company'].unique() # To show companies
```

```
Out[7]: array(['OpenAI', 'Google', 'Meta'], dtype=object)
```

```
In [8]: data['Year'] = data['Date'].dt.year # To add Year column
```

```
In [9]: data.head(2)
```

```
Out[9]:
```

	Date	Company	R&D_Spending_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth_%
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0	2015-01-01	OpenAI	5.92	0.63	-36.82
1	2015-01-02	OpenAI	5.41	1.81	80.59



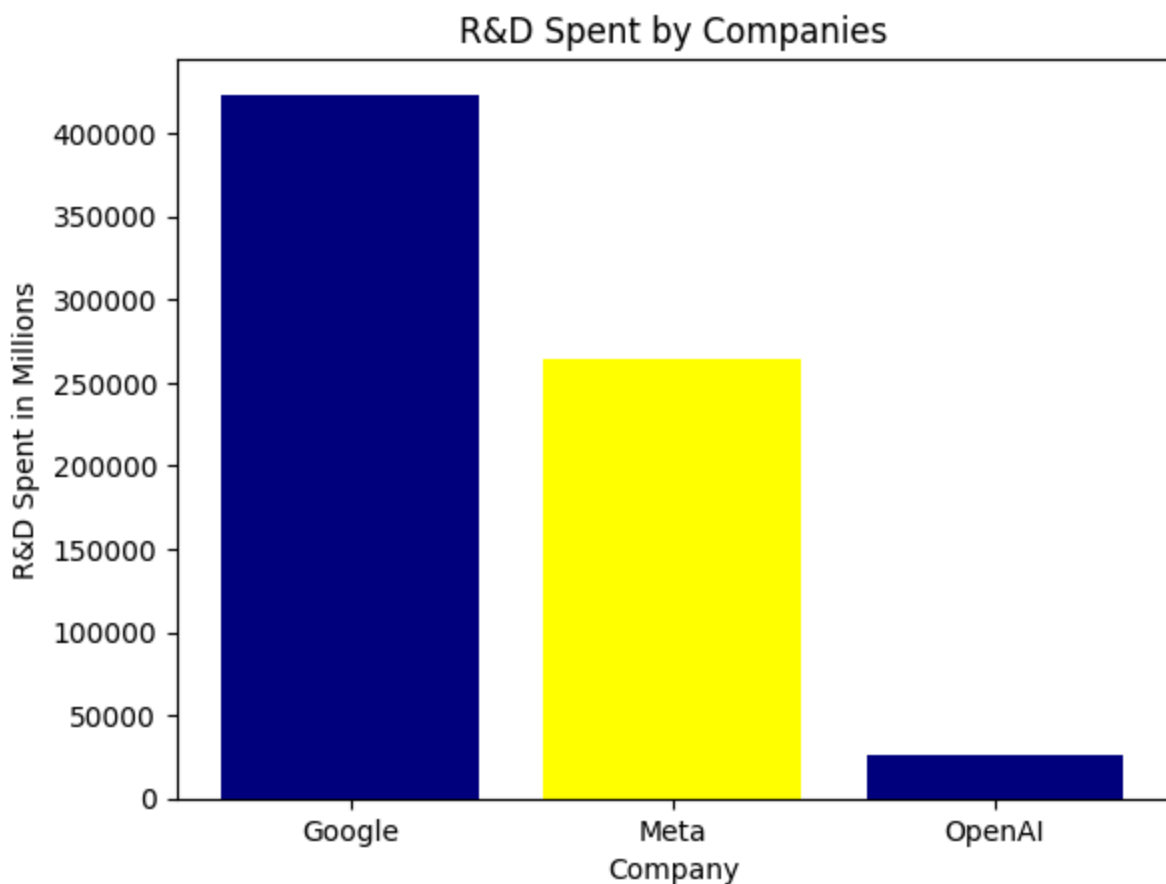
1. How much Companies spent on R&D ?

```
In [10]: print('Companies spending on R&D')
rd = data.groupby('Company')['R&D_Spending_USD_Mn'].sum()
rd
```

Companies spending on R&D

```
Out[10]: Company
Google    423341.14
Meta      264533.07
OpenAI     26482.77
Name: R&D_Spending_USD_Mn, dtype: float64
```

```
In [11]: plt.bar( rd.index, rd.values , color = ['navy','yellow'])
plt.title('R&D Spent by Companies')
plt.xlabel('Company')
plt.ylabel('R&D Spent in Millions')
plt.show();
```



2. Revenue earned by the Companies

```
In [12]: data.head(2)
```

Out[12]:

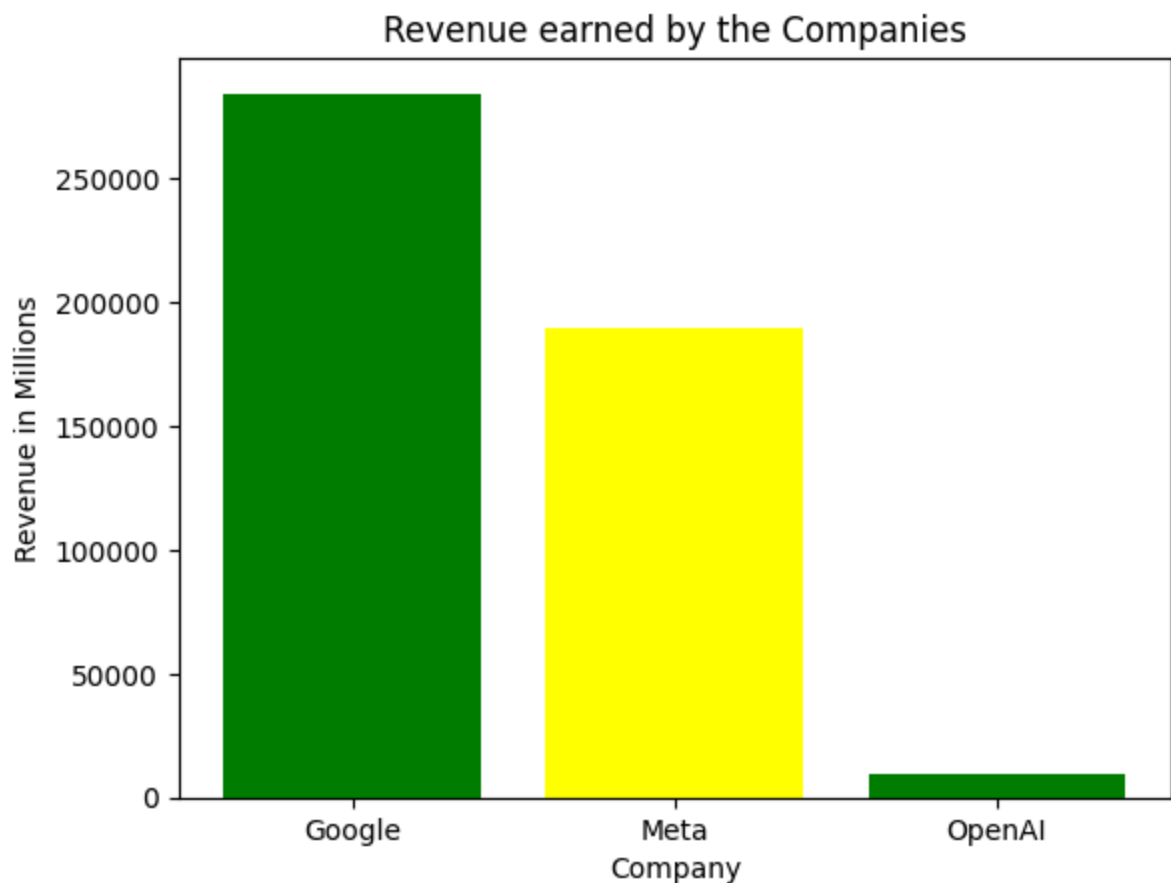
	Date	Company	R&D_Spending_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth_%
0	2015-01-01	OpenAI	5.92	0.63	-36.82
1	2015-01-02	OpenAI	5.41	1.81	80.59



In [13]: `RE = data.groupby('Company')['AI_Revenue_USD_Mn'].sum()`
`RE`

Out[13]: Company
 Google 284498.38
 Meta 189621.82
 OpenAI 9462.89
 Name: AI_Revenue_USD_Mn, dtype: float64

In [14]: `plt.bar(RE.index, RE.values, color = ['green', 'yellow'])`
`plt.title('Revenue earned by the Companies')`
`plt.xlabel('Company')`
`plt.ylabel('Revenue in Millions')`
`plt.show();`

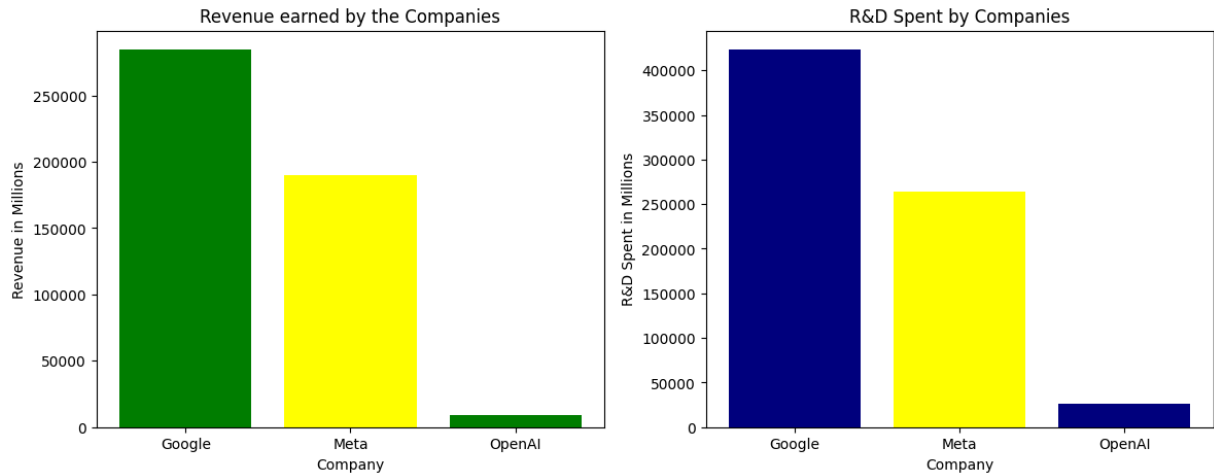


In [15]: `plt.figure(figsize = (14,5))`
`plt.subplot(1,2,1)`

```
plt.bar(RE.index, RE.values, color = ['green', 'yellow'])
plt.title('Revenue earned by the Companies')
plt.xlabel('Company')
plt.ylabel('Revenue in Millions')

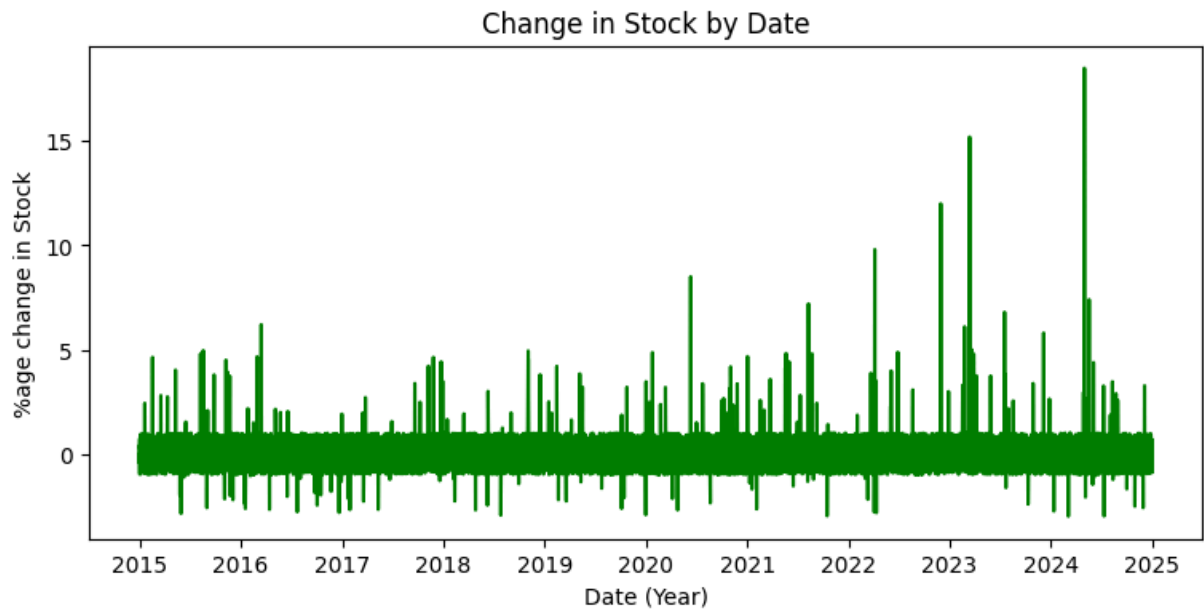
plt.subplot(1,2,2)

plt.bar( rd.index, rd.values , color = ['navy','yellow'])
plt.title('R&D Spent by Companies')
plt.xlabel('Company')
plt.ylabel('R&D Spent in Millions')
plt.show();
```



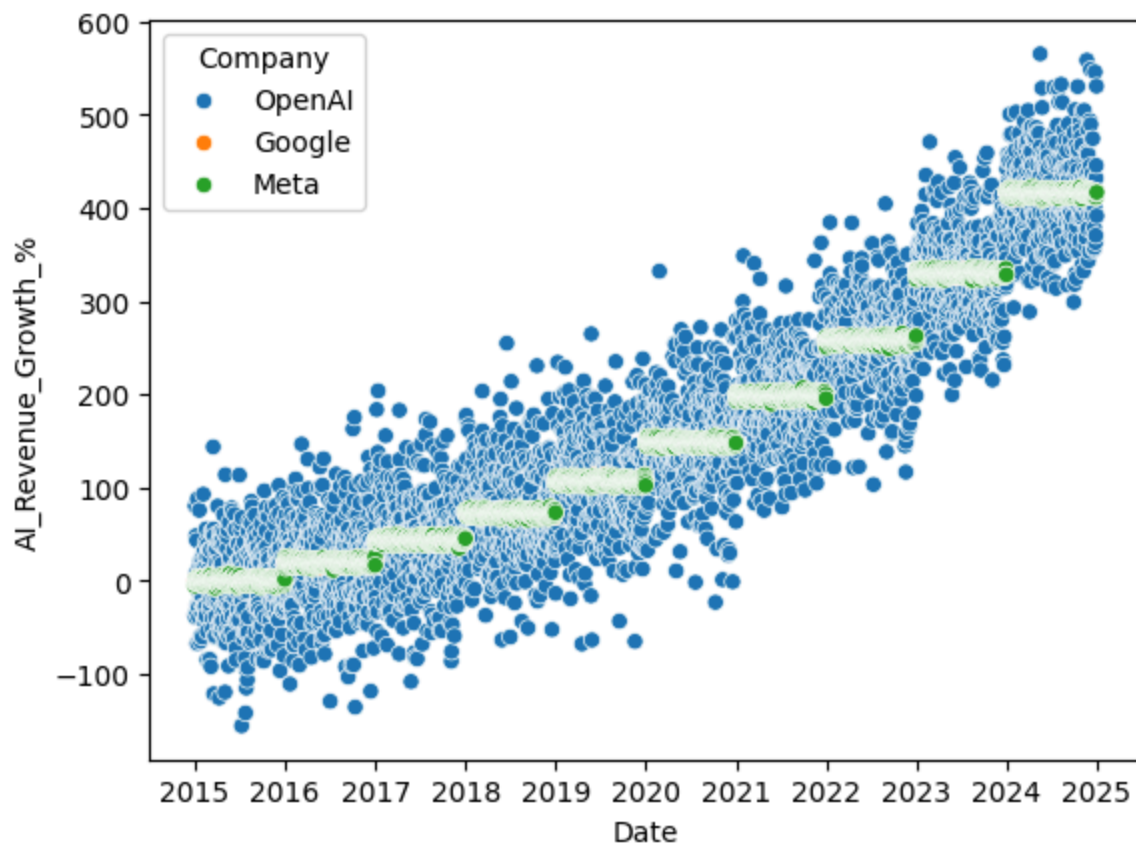
3. Datewise impact on Stock

```
In [16]: plt.figure(figsize = (9,4))
plt.plot(data['Date'], data['Stock_Impact_%'], color = 'green')
plt.title('Change in Stock by Date')
plt.xlabel('Date (Year)')
plt.ylabel('%age change in Stock')
plt.show();
```



4. AI Revenue growth of Companies

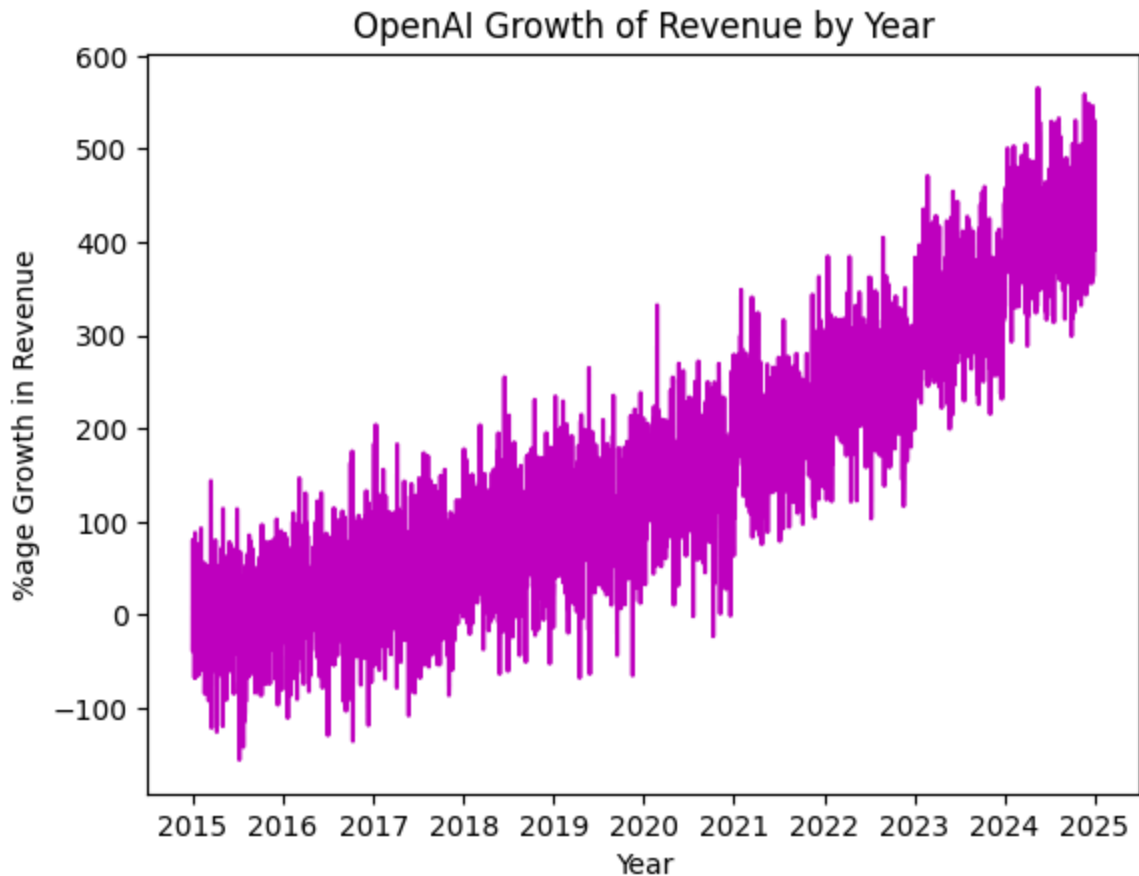
```
In [17]: sns.scatterplot(x = data['Date'], y = data['AI_Revenue_Growth_%'], data= data, hue
plt.show()
```



5. Open AI's Growth by Year

```
In [18]: open_ai = data[data['Company'] == 'OpenAI']
```

```
In [19]: plt.plot(open_ai['Date'], open_ai['AI_Revenue_Growth_%'], color = 'm')
plt.title('OpenAI Growth of Revenue by Year')
plt.xlabel('Year')
plt.ylabel('%age Growth in Revenue')
plt.show();
```



6. Correlation between the Columns

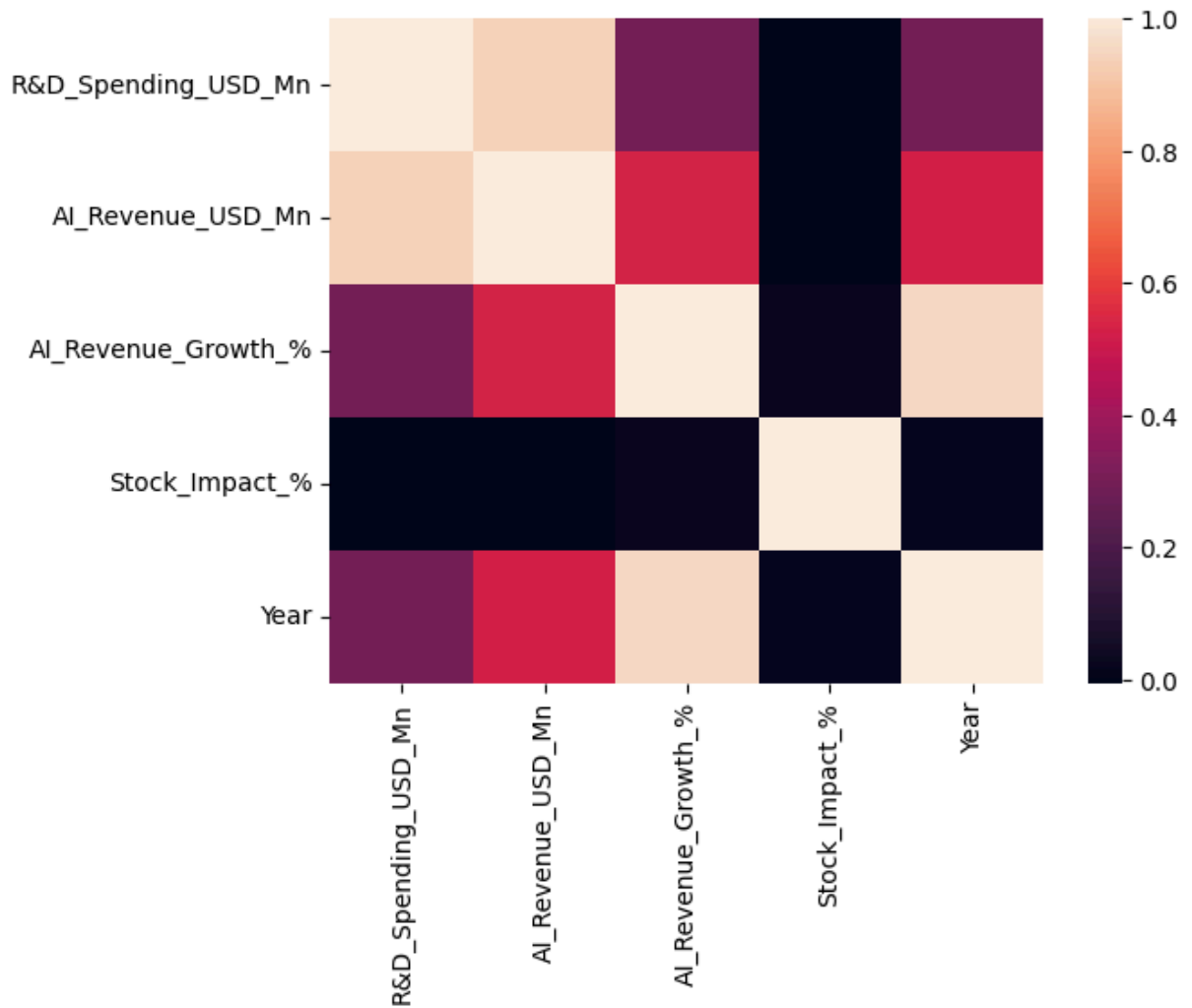
```
In [20]: data.corr(numeric_only = True)
```

```
Out[20]:
```

	R&D_Spending_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth_
R&D_Spending_USD_Mn	1.000000	0.935030	0.2912
AI_Revenue_USD_Mn	0.935030	1.000000	0.5301
AI_Revenue_Growth_	0.291245	0.530128	1.0000
Stock_Impact_	-0.004628	-0.000017	0.0209
Year	0.297701	0.526478	0.9521

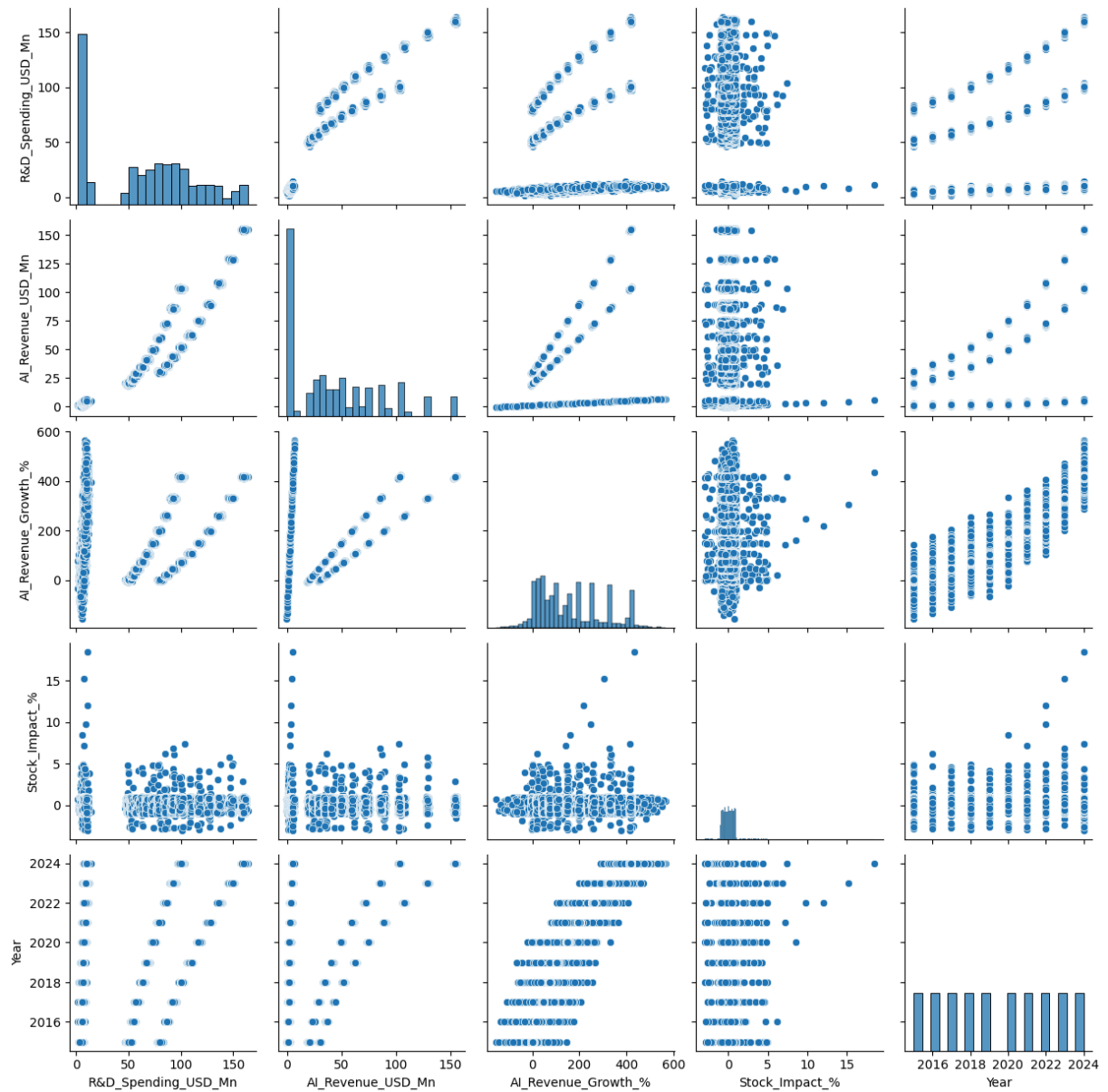


```
In [21]: sns.heatmap(data.corr(numeric_only = True));
```



7. Draw Pairplot to show different relations between columns

```
In [22]: sns.pairplot(data);
```

8. Event Impact Analysis

```
In [23]: data.Event.value_counts()
```

```
Out[23]: Event
AI speech recognition release    37
AI Ads Optimization upgrade      32
AI partnership deal              32
AI-powered search update         32
AI Video Recommendation upgrade  29
AI ethics policy update          27
Cloud AI launch                  27
GPT-1 release                    1
GPT-2 release                    1
GPT-3 release                    1
Codex release                    1
DALL·E 2 release                 1
ChatGPT (GPT-3.5) launch        1
GPT-4 release                    1
GPT-5 release (predicted)       1
TensorFlow open-source release  1
AlphaGo beats Lee Sedol         1
BERT for Search launch          1
MUM Search Model launch         1
Bard chatbot launch             1
Gemini AI release               1
LLaMA 1 release                  1
LLaMA 2 release                  1
LLaMA 3 release (predicted)     1
Name: count, dtype: int64
```

```
In [24]: data[data['Event']=='LLaMA 1 release']
```

```
Out[24]:
```

	Date	Company	R&D_Spending_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth
--	------	---------	---------------------	-------------------	-------------------

10282	2023-02-24	Meta	93.71	86.98	334
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```
In [25]: ef = data.loc[10272 : 10292 ]
ef
```

Out[25]:

	Date	Company	R&D_Spending_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth
10272	2023-02-14	Meta	92.51	85.57	327
10273	2023-02-15	Meta	93.65	86.11	330
10274	2023-02-16	Meta	93.38	86.29	331
10275	2023-02-17	Meta	93.92	85.65	328
10276	2023-02-18	Meta	92.72	86.97	334
10277	2023-02-19	Meta	92.64	86.65	333
10278	2023-02-20	Meta	92.69	85.61	328
10279	2023-02-21	Meta	93.43	86.16	330
10280	2023-02-22	Meta	93.28	86.32	331
10281	2023-02-23	Meta	90.92	85.67	328
10282	2023-02-24	Meta	93.71	86.98	334
10283	2023-02-25	Meta	93.44	86.02	330
10284	2023-02-26	Meta	93.71	85.53	327
10285	2023-02-27	Meta	93.19	86.53	332
10286	2023-02-28	Meta	93.42	85.82	329
10287	2023-03-01	Meta	93.09	87.11	335
10288	2023-03-02	Meta	93.27	86.12	330
10289	2023-03-03	Meta	93.18	85.93	329
10290	2023-03-04	Meta	93.01	85.81	329

	Date	Company	R&D_Spending_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth
10291	2023-03-05	Meta	91.77	86.38	331
10292	2023-03-06	Meta	91.71	86.25	331

```
In [27]: plt.figure(figsize = (9,4))
plt.plot(ef['Date'], ef['Stock_Impact_%'], color = 'c')
plt.show();
```



9. Change the index WRT Company & Year

```
In [28]: data.head(2)
```

```
Out[28]:
```

	Date	Company	R&D_Spending_USD_Mn	AI_Revenue_USD_Mn	AI_Revenue_Growth_%
0	2015-01-01	OpenAI	5.92	0.63	-36.82
1	2015-01-02	OpenAI	5.41	1.81	80.59



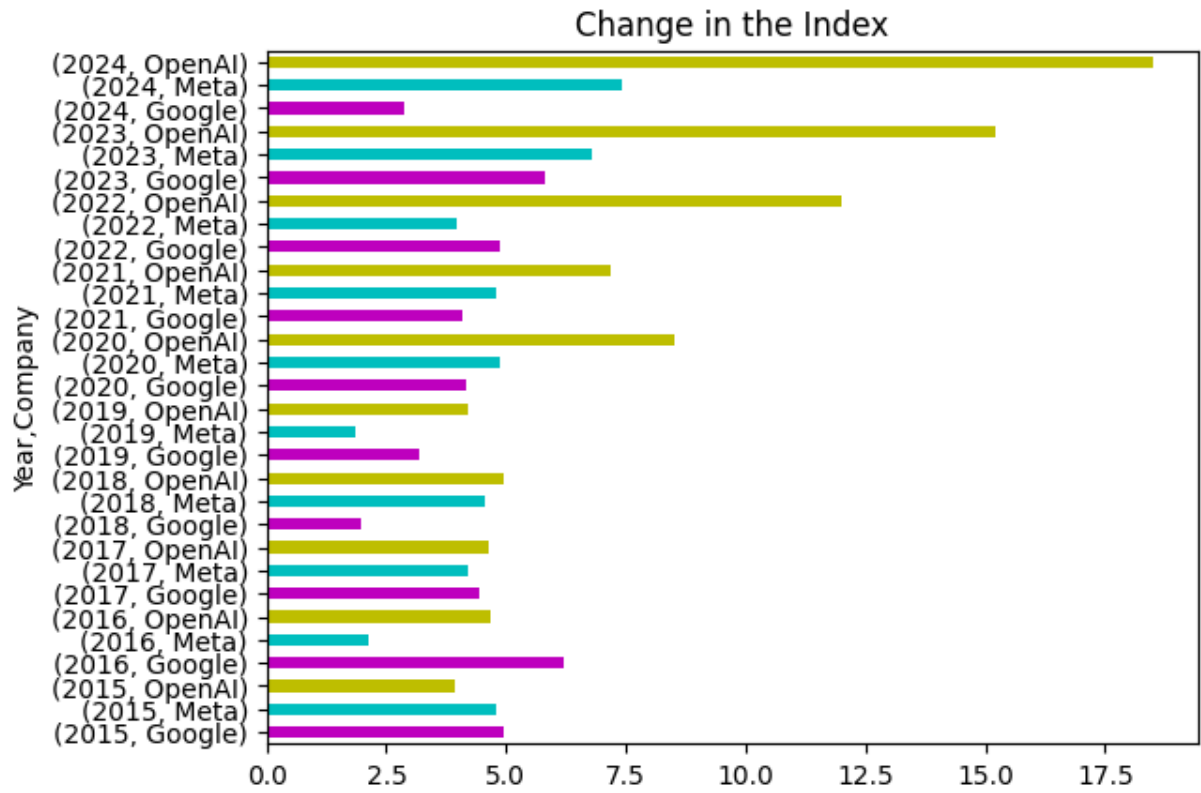
```
In [31]: stocks = data.groupby(['Year', 'Company'])['Stock_Impact_%'].max()
stocks
```

```
Out[31]:
```

Year	Company	
2015	Google	4.96
	Meta	4.80
	OpenAI	3.92
2016	Google	6.20
	Meta	2.14
	OpenAI	4.66
2017	Google	4.42
	Meta	4.21
	OpenAI	4.63
2018	Google	1.97
	Meta	4.54
	OpenAI	4.94
2019	Google	3.20
	Meta	1.86
	OpenAI	4.20
2020	Google	4.18
	Meta	4.86
	OpenAI	8.50
2021	Google	4.10
	Meta	4.80
	OpenAI	7.20
2022	Google	4.88
	Meta	3.98
	OpenAI	12.00
2023	Google	5.80
	Meta	6.80
	OpenAI	15.20
2024	Google	2.89
	Meta	7.40
	OpenAI	18.50

Name: Stock_Impact_%, dtype: float64

```
In [34]: stocks.plot(kind = 'barh', color = ['m','c','y'])  
plt.title('Change in the Index')  
plt.show();
```



In []: